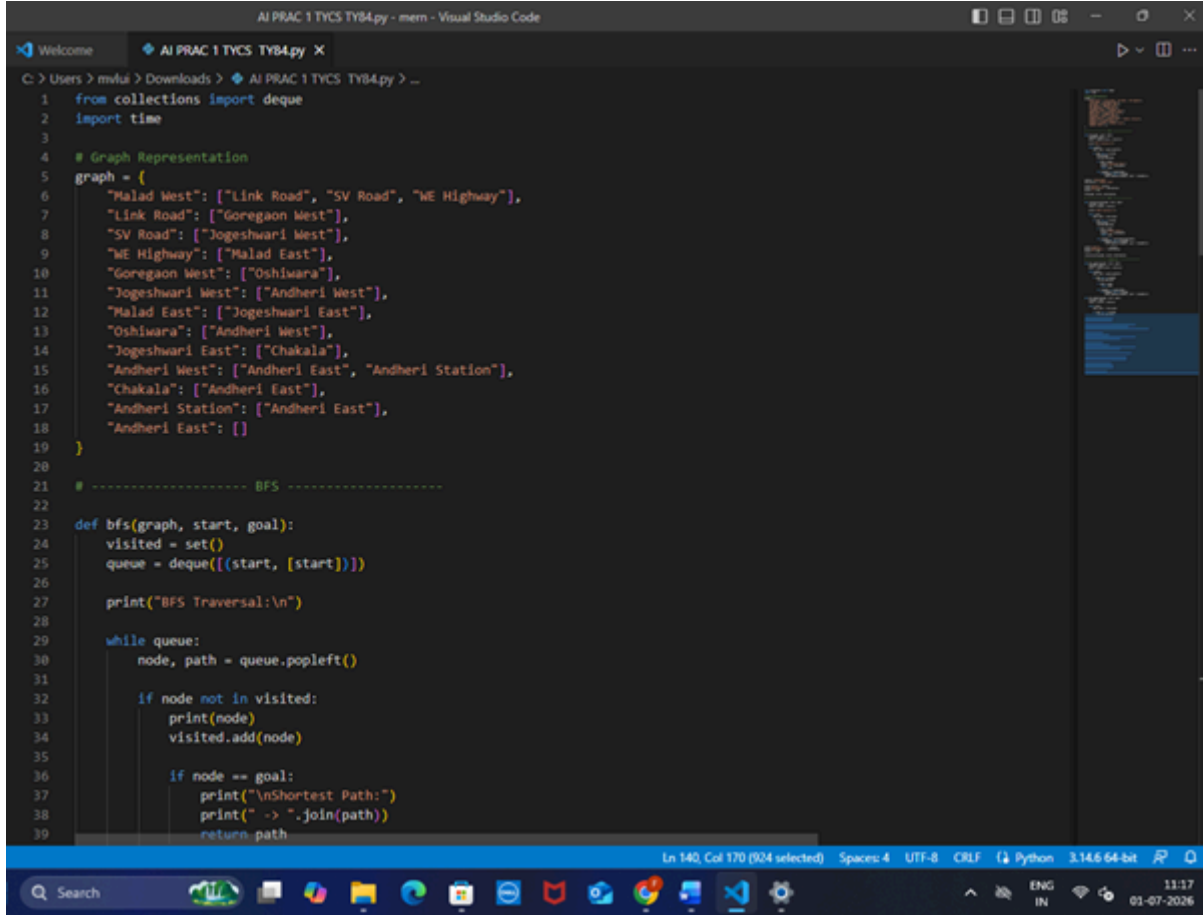


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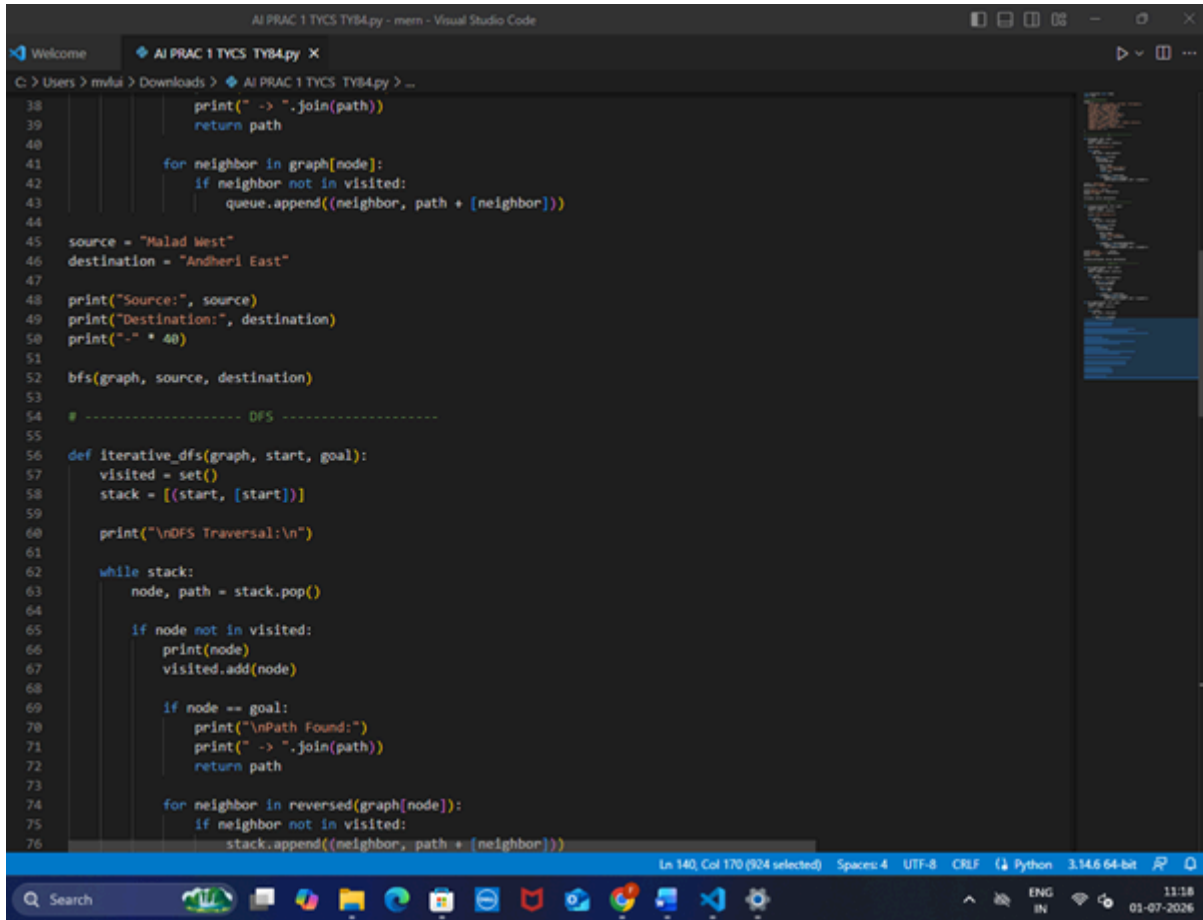
INPUT :-



```
AI PRAC 1 TYCS TYB4.py - mem - Visual Studio Code
Welcome AI PRAC 1 TYCS TYB4.py X
C: > Users > mful > Downloads > AI PRAC 1 TYCS TYB4.py > ...
1 from collections import deque
2 import time
3
4 # Graph Representation
5 graph = {
6     "Malad West": ["Link Road", "SV Road", "WE Highway"],
7     "Link Road": ["Goregaon West"],
8     "SV Road": ["Jogeshwari West"],
9     "WE Highway": ["Malad East"],
10    "Goregaon West": ["Oshiwara"],
11    "Jogeshwari West": ["Andheri West"],
12    "Malad East": ["Jogeshwari East"],
13    "Oshiwara": ["Andheri West"],
14    "Jogeshwari East": ["Chakala"],
15    "Andheri West": ["Andheri East", "Andheri Station"],
16    "Chakala": ["Andheri East"],
17    "Andheri Station": ["Andheri East"],
18    "Andheri East": []
19 }
20
21 # ----- BFS -----
22
23 def bfs(graph, start, goal):
24     visited = set()
25     queue = deque([(start, [start])])
26
27     print("BFS Traversal:\n")
28
29     while queue:
30         node, path = queue.popleft()
31
32         if node not in visited:
33             print(node)
34             visited.add(node)
35
36             if node == goal:
37                 print("\nShortest Path:")
38                 print(" -> ".join(path))
39                 return path
```

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ROLL NO :- TYCS84

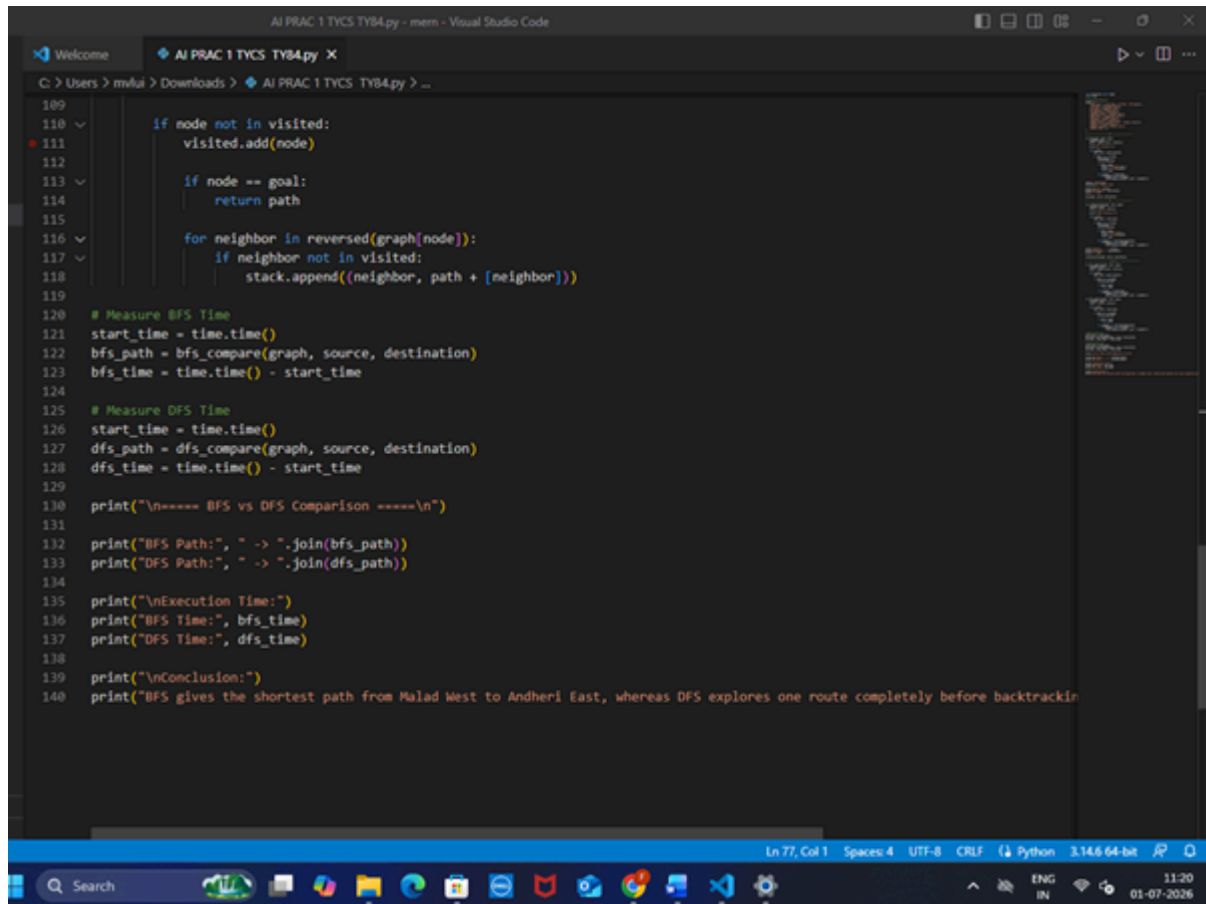
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```
38         print(" -> ".join(path))
39         return path
40
41     for neighbor in graph[node]:
42         if neighbor not in visited:
43             queue.append((neighbor, path + [neighbor]))
44
45     source = "Malad West"
46     destination = "Andheri East"
47
48     print("Source:", source)
49     print("Destination:", destination)
50     print("-" * 40)
51
52     bfs(graph, source, destination)
53
54     # ----- DFS -----
55
56     def iterative_dfs(graph, start, goal):
57         visited = set()
58         stack = [(start, [start])]
59
60         print("\nDFS Traversal:\n")
61
62         while stack:
63             node, path = stack.pop()
64
65             if node not in visited:
66                 print(node)
67                 visited.add(node)
68
69                 if node == goal:
70                     print("\nPath Found:")
71                     print(" -> ".join(path))
72                     return path
73
74                 for neighbor in reversed(graph[node]):
75                     if neighbor not in visited:
76                         stack.append((neighbor, path + [neighbor]))
```

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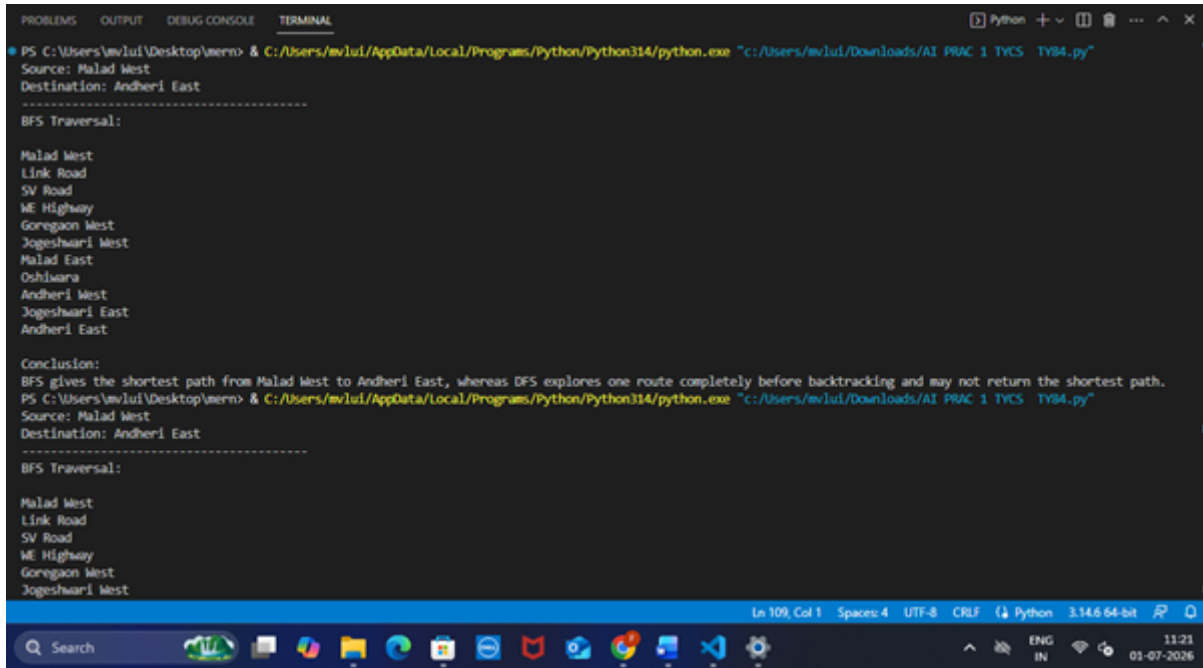


```
109
110     if node not in visited:
111         visited.add(node)
112
113         if node == goal:
114             return path
115
116         for neighbor in reversed(graph[node]):
117             if neighbor not in visited:
118                 stack.append((neighbor, path + [neighbor]))
119
120 # Measure BFS Time
121 start_time = time.time()
122 bfs_path = bfs_compare(graph, source, destination)
123 bfs_time = time.time() - start_time
124
125 # Measure DFS Time
126 start_time = time.time()
127 dfs_path = dfs_compare(graph, source, destination)
128 dfs_time = time.time() - start_time
129
130 print("\n===== BFS vs DFS Comparison =====\n")
131
132 print("BFS Path:", " -> ".join(bfs_path))
133 print("DFS Path:", " -> ".join(dfs_path))
134
135 print("\nExecution Time:")
136 print("BFS Time:", bfs_time)
137 print("DFS Time:", dfs_time)
138
139 print("\nConclusion:")
140 print("BFS gives the shortest path from Malad West to Andheri East, whereas DFS explores one route completely before backtracking")
```

OUTPUT :-

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```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\mvlui\Desktop\mern> & C:\Users\mvlui\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/mvlui/Downloads/AI PRAC 1 TYCS TY84.py"
Source: Malad West
Destination: Andheri East
-----
BFS Traversal:

Malad West
Link Road
SV Road
ME Highway
Goregaon West
Jogeshwari West
Malad East
Oshiwara
Andheri West
Jogeshwari East
Andheri East

Conclusion:
BFS gives the shortest path from Malad West to Andheri East, whereas DFS explores one route completely before backtracking and may not return the shortest path.
PS C:\Users\mvlui\Desktop\mern> & C:\Users\mvlui\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/mvlui/Downloads/AI PRAC 1 TYCS TY84.py"
Source: Malad West
Destination: Andheri East
-----
BFS Traversal:

Malad West
Link Road
SV Road
ME Highway
Goregaon West
Jogeshwari West
```

Ln 109, Col 1 Spaces: 4 UTF-8 CRLF Python 3.14.6 64-bit

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